

Room: <https://tryhackme.com/room/splunk101>
Room Name: **Splunk: Basics**

Topic 3: Splunk Components

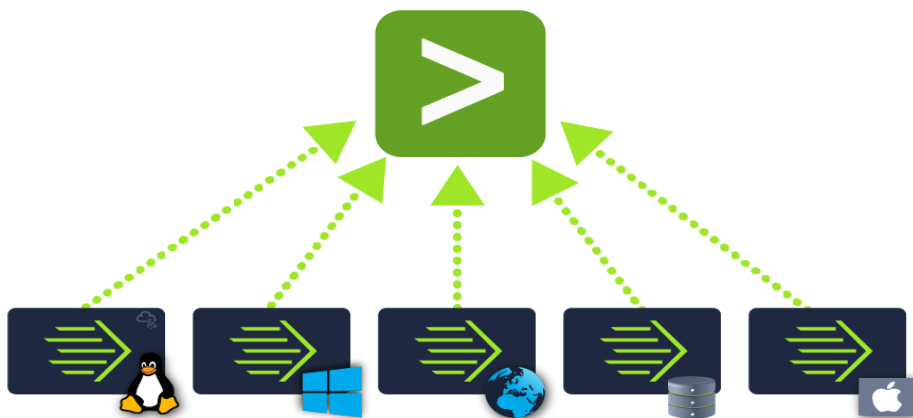
Splunk has three main components, namely Forwarder, Indexer, and Search Head. These components are explained below:



Splunk Forwarder

Splunk Forwarder is a lightweight agent installed on the endpoint intended to be monitored, and its main task is to collect the data and send it to the Splunk instance. It does not affect the endpoint's performance, as it takes very few resources to process. Some of the key data sources are:

- Web server generating web traffic.
- Windows machine generating Windows Event Logs, PowerShell, and Sysmon data.
- Linux host generating host-centric logs.
- Database generating DB connection requests, responses, and errors.



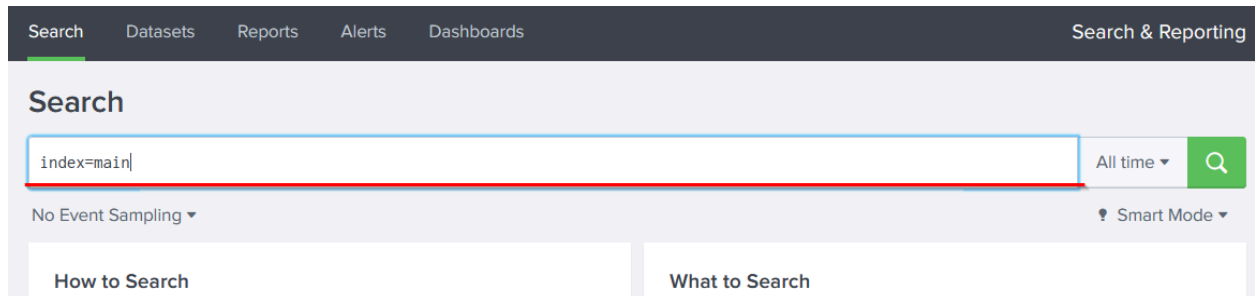
Splunk Indexer

The Splunk Indexer plays the main role in processing the data it receives from forwarders. It takes the data, normalizes it into field-value pairs, determines the datatype of the data, and stores it as events. Processed data is easy to search and analyze.



Search Head

Splunk Search Head is the place within the Search & Reporting App where users can search the indexed logs, as shown below. When the user searches for a term or uses a Search language known as Splunk Search Processing Language, the request is sent to the indexer, and the relevant events are returned in the form of field-value pairs.



Search Head also provides the ability to transform the results into presentable tables and visualizations like pie charts, bar charts, and column charts, as shown below:

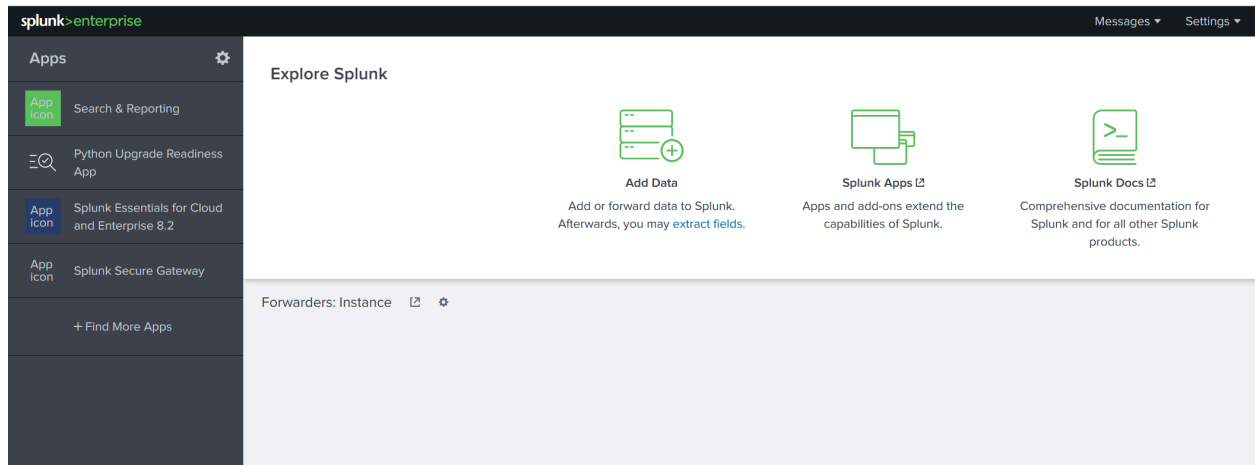
Qn1: Which component is used to collect and send data over the Splunk instance?

Answer: Forwarder

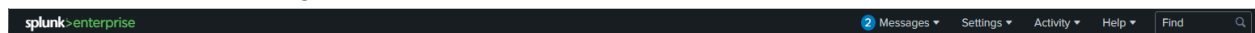
Topic 4: Navigating Splunk

Splunk Bar

When you access Splunk, you will see the default home screen identical to the screenshot below.

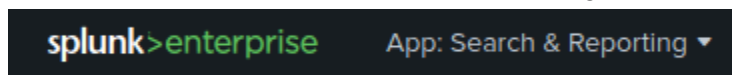


Let's look at each section, or panel, that makes up the home screen. The top panel is the Splunk Bar (below image).



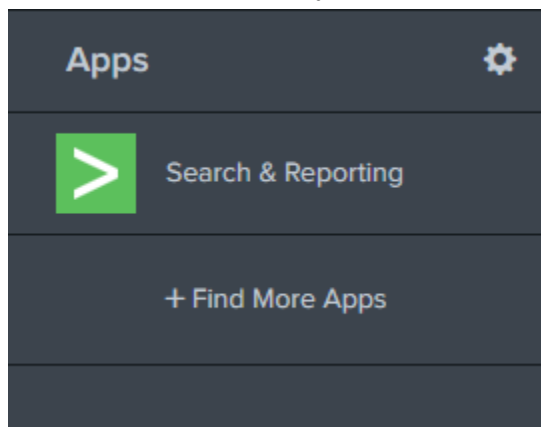
In the Splunk Bar, you can see system-level messages (Messages), configure the Splunk instance (Settings), review the progress of jobs (Activity), find miscellaneous information such as tutorials (Help), and use a search feature (Find).

The ability to switch between installed Splunk apps instead of using the Apps panel can be achieved from the Splunk Bar, like in the image below.



Apps Panel

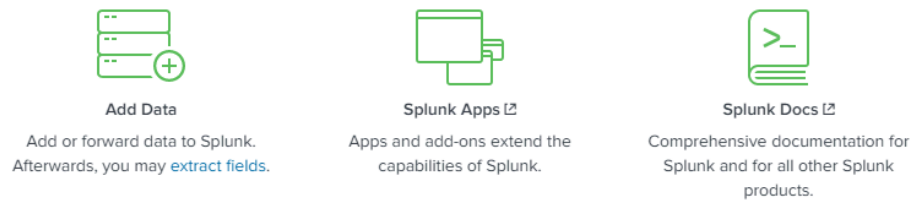
Next is the Apps Panel. In this panel, you can see the apps installed for the Splunk instance. The default app for every Splunk installation is Search & Reporting.



Explore Splunk:

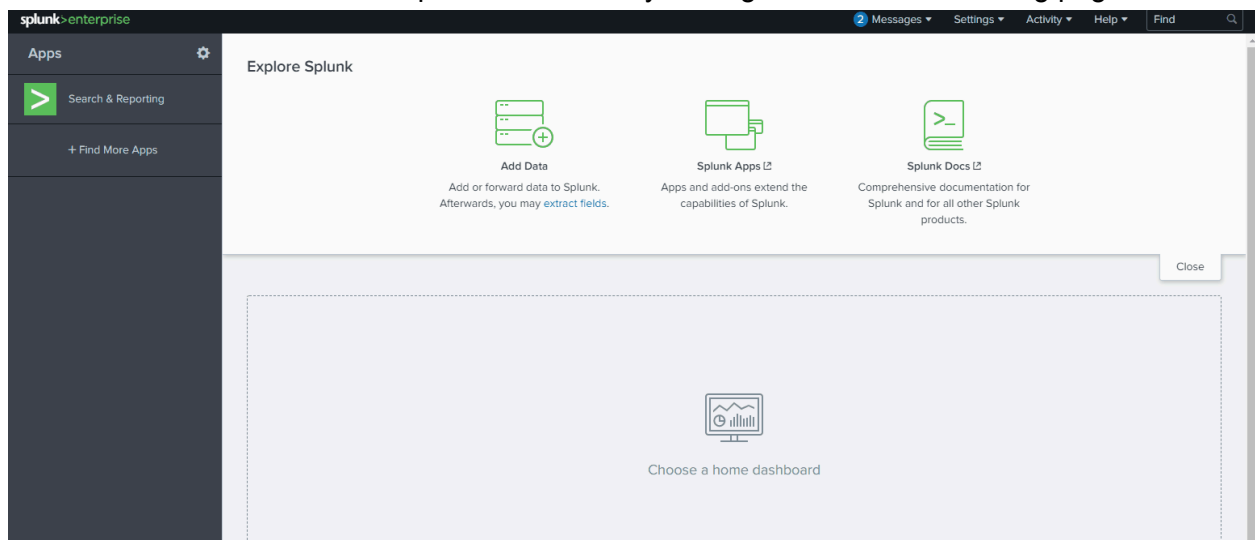
The next section is Explore Splunk. This panel contains quick links to add data to the Splunk instance, add new Splunk apps, and access the Splunk documentation.

Explore Splunk



Splunk Dashboard

The last section is the Home Dashboard. By default, no dashboards are displayed. You can choose from a range of dashboards readily available within your Splunk instance. You can select a dashboard from the dropdown menu or by visiting the dashboards listing page .



You can also create dashboards and add them to the Home Dashboard. The dashboards you create can be viewed isolated from the other dashboards by clicking on the "Yours" tab.

Qn1: In the Add Data tab, which option is used to collect data from files and ports?

Answer: Monitor

Topic 5: Adding Data

For adding Data into Splunk, I have done this.

The screenshot displays the Splunk 'Add Data' workflow. At the top, the 'Add Data' button is highlighted with a red circle and arrow labeled '1'. Below this, the 'Forwarders: Instance' section is visible. The main area shows 'Follow guides for onboarding popular data sources' with categories like Cloud computing, Networking, Operating System, and Security. Below these, 'Or get data in with the following methods' shows 'Upload', 'Monitor', and 'Forward'. The 'Upload' method is selected, indicated by a red circle and arrow labeled '1'. The 'Add Data' progress bar at the top shows 'Select Source' as the current step, with 'Next >' highlighted by a red circle and arrow labeled '3'. The 'Select Source' screen shows 'Selected File: VPNlogs.json' and a 'Select File' button, which is highlighted by a red circle and arrow labeled '2'. A large red circle and arrow labeled '1' points to the 'Drop your data file here' area.

Explore Splunk

Forwarders: Instance

Forwarder Monitoring is disabled. Please go to the [forwarder](#) page to enable it.

what data do you want to send to the Splunk platform?

Follow guides for onboarding popular data sources

Cloud computing
Get your cloud computing data in to the Splunk platform.
10 data sources

Networking
Get your networking data in to the Splunk platform.
2 data sources

Operating System
Get your operating system data in to the Splunk platform.
1 data source

Security
Get your security data in to the Splunk platform.
3 data sources

4 data sources in total

Or get data in with the following methods

Upload
files from my computer
Local log files
Local structured files (e.g., CSV)
[Tutorial for adding data](#)

Monitor
files and ports on this Splunk platform instance
Files - HTTP - WMI - TCP/UDP - Scripts
Modular inputs for external data sources

Forward
data from a Splunk forwarder
Files - TCP/UDP - Scripts

Add Data

Select Source Set Source Type Input Settings Review Done

< Back Next >

Select Source

Choose a file to upload to the Splunk platform, either by browsing your computer or by pasting a URL into the target box below. [Learn More](#)

Selected File: VPNlogs.json

Select File

Drop your data file here

The maximum file upload size is 500 Mb

Done

We have to select our Log files.

Add Data

Select SourceSet Source TypeInput SettingsReviewDone

< Back

Next >

1

Add Data

Select SourceSet Source TypeInput SettingsReviewDone

< Back

Review >

Input Settings

Optionally set additional input parameters for this data input as follows:

Host

When the Splunk platform indexes data, each event receives a "host" value. The host value should be the name of the machine from which the event originates. The type of input you choose determines the available configuration options. [Learn More](#)

☒ Constant value
☐ Regular expression on path
☐ Segment in path

Host field value

Index

The Splunk platform stores incoming data as events in the selected index. Consider using a "sandbox" index as a destination if you have problems determining a source type for your data. A sandbox index lets you troubleshoot your configuration without impacting production indexes. You can always change this setting later. [Learn More](#)

Index Default ▾ [Create a new index](#)

FAQ

> How do indexes work?

1

Add Data

Select SourceSet Source TypeInput SettingsReviewDone

< Back

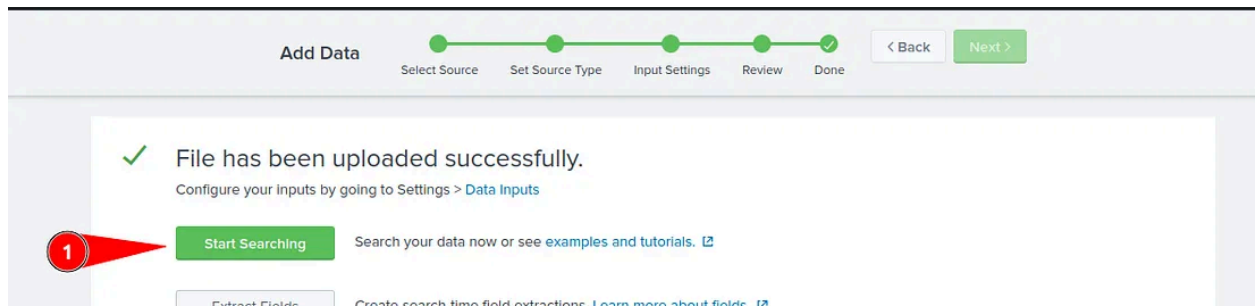
Submit >

Review

Input Type Uploaded File
File Name VPNiogs.json
Source Type _json
Host ip-10-10-40-195
Index Default

1

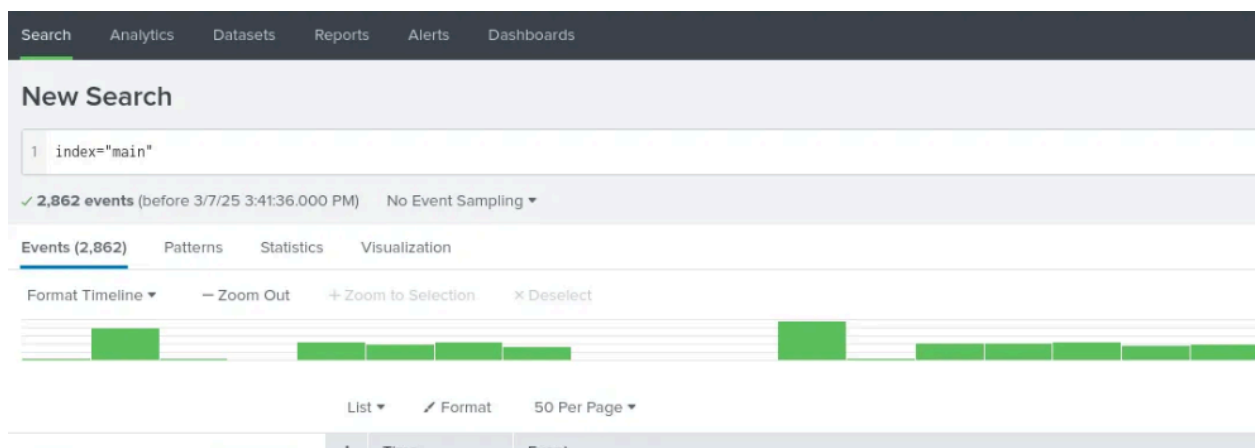
After submitting the Data, we can analyze it from the search



Qn1: Upload the data attached to this task and create an index "VPN_Logs." How many events are present in the log file?

Answer: 2862

Input: **index="main"** [this will show all the logs under index main]



Qn2 How many log events by the user Maleena are captured?

Answer: 60 [From the username FIELD I need to filter the user 'Maleena.']

Format Timeline ▾ — Zoom Out + Zoom to Selection × Deselect

List ▾ ↗ Format 50 Per Page ▾

< Hide Fields ≡ All Fields

SELECTED FIELDS

- a host 1
- a source 1
- a Source_Country 7
- a sourcetype 1

INTERESTING FIELDS

- a action 3
- a Company 1
- # date_hour 23
- # date_mday 27
- # date_minute 60
- a date_month 2
- # date_second 60
- a date_wday 7
- # date_year 2
- a date_zone 1
- a EventTime 100+
- a extracted_index 1
- a index 1
- # linecount 1
- # port 1
- a protocol 1
- a punct 4
- a Source_ip 100+
- a source_state 16
- a splunk_server 1
- # timeendpos 19
- # timestartpos 23
- a Username 51

Time **Event**

> 1/31/22 6:29:41.000 PM { [-]

Company: CyberT
EventTime: 2022-01-31T18:29:41
Source_Country: United States
Source_ip: 143.23.45.158
Username: Phill
action: teardown
index: VPN_Logs
port: 443
protocol: tcp

Username

51 Values, 100% of events Selected Yes No

Reports

Top values Top values by time Rare values

Events with this field

Top 10 Values	Count	%
Simon	278	9.713%
James	108	3.774%
Maleena	60	2.096%
Rock	60	2.096%
Bentle	58	2.026%
Paul King	58	2.026%
Emanda	56	1.957%
Kate Wistle	56	1.957%

New Search

1 index="main" UserName=Maleena

✓ 60 events (before 3/7/25 3:43:13.000 PM) No Event Sampling ▼

Events (60) Patterns Statistics Visualization

Format Timeline ▼ — Zoom Out + Zoom to Selection × Deselect

List ▼ Format 50 Per Page ▼

Hide Fields	All Fields	i	Time	Event
SELECTED FIELDS		>	1/31/22 4:33:56.000 PM	{ [-] Company: CyberT EventTime: 2022-01-31T16:33:56 Source_Country: United States Source_ip: 162.148.170.20 UserName: Maleena action: teardown

INTERESTING FIELDS

Here is the Answer
60

Qn3 What is the name associated with IP 107.14.182.38?

ANS: Input index="main" Source_ip="107.14.182.38" [into the search command, I have filtered the source IP: 107.14.182.38]

New Search

1 index="main" Source_ip="107.14.182.38"

✓ 26 events (before 3/7/25 3:44:25.000 PM) No Event Sampling ▼

Events (26) Patterns Statistics Visualization

Format Timeline ▼ — Zoom Out + Zoom to Selection × Deselect

List ▼ Format 50 Per Page ▼

Hide Fields	All Fields	i	Time	Event
SELECTED FIELDS		>	1/31/22 6:22:08.000 PM	{ [-] Company: CyberT EventTime: 2022-01-31T18:22:08 Source_Country: United States Source_ip: 107.14.182.38 UserName: Smith action: teardown index: VPN_Logs port: 443 protocol: tcp source_state: Tennessee

INTERESTING FIELDS

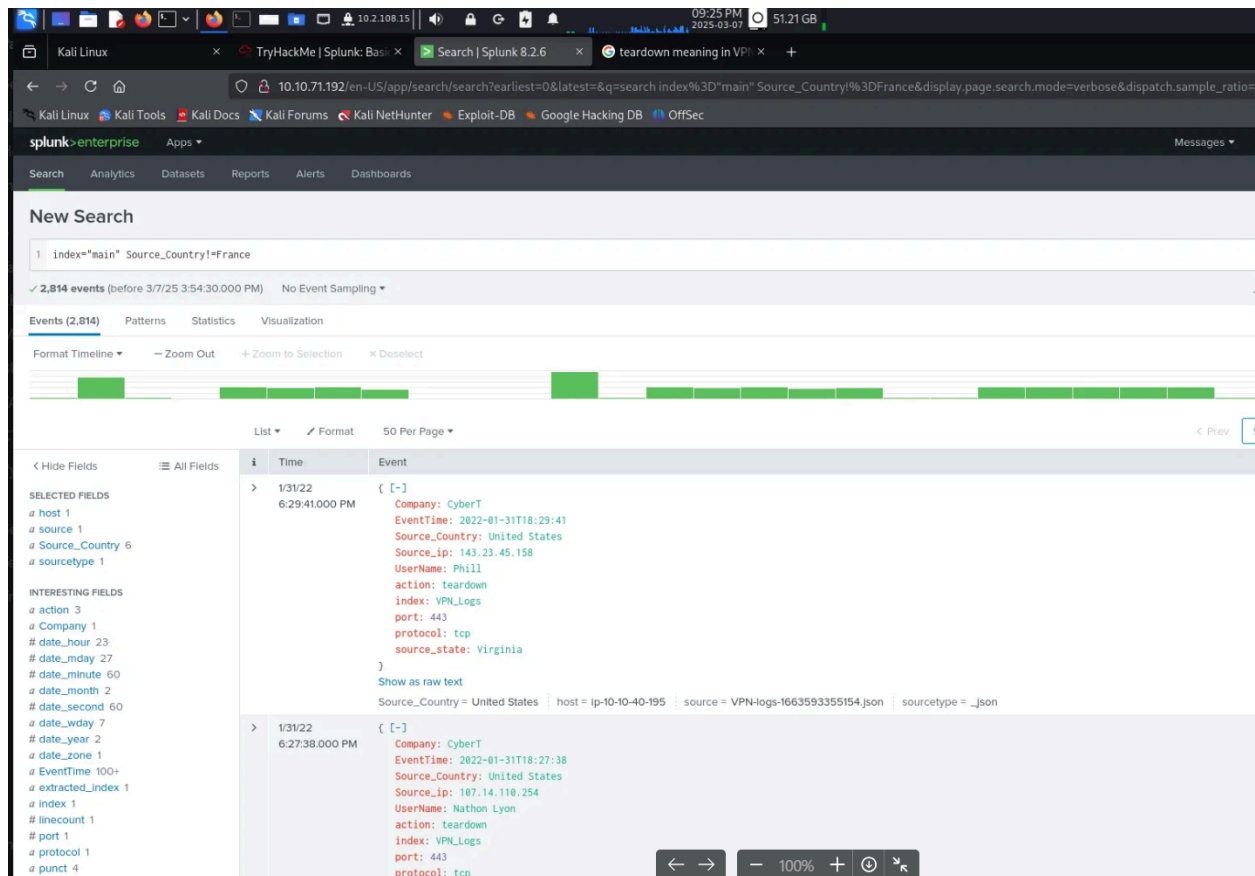
Source_Country = United States | host = ip-10-10-40-195 | source = VPN-logs-1663593355154.json | sourcetype = _json

Answer: Smith

Qn4: What is the number of events that originated from all countries except France?

Answer:

Input: `index="main" Source_Country!=France` [it will filter the source countries except France because of this '!']



Here is the Answer: 2814

Qn5: How many VPN Events were observed by the IP 107.3.206.58?

Answer:

Input: Index="main" Source_ip="107.3.206.58"

The screenshot shows the Splunk Enterprise web interface. The search bar contains the query: `index="main" Source_ip="107.3.206.58"`. Below the search bar, it indicates 14 events were found. The results are displayed in a table format with columns for Time and Event. The event details show a VPN log entry for a user named Will Smith, with fields like Company, EventTime, Source_Country, Source_Id, Username, action, index, port, protocol, and source_state.

Time	Event
1/31/22 5:50:18.000 PM	<pre>{ "Company": "CyberT", "EventTime": "2022-01-31T17:50:18", "Source_Country": "United States", "Source_Id": "107.3.206.58", "Username": "Will Smith", "action": "teardown", "index": "VPN_Logs", "port": 443, "protocol": "tcp", "source_state": "Virginia" }</pre>

Output is 14.

The screenshot shows the TryHackMe interface for the 'Splunk: Basics' room. A green notification box says 'Woop woop! Your answer is correct'. The main message reads 'You did it! 🎉 Splunk: Basics complete!'. Below this, statistics are shown: Points earned (56), Completed tasks (6), Room type (Walkthrough), Difficulty (Easy), and Streak (6). At the bottom, it states '72,531 users are actively learning this week' and provides buttons for 'Leave Feedback' and 'Continue'.